

## Quiz - 4

- ① Find the derivative of the distribution:

$$f(t) = \sin(t) u(\pi/2 - t)$$

- ② Sketch the function

$$f(t) = \int_{-\infty}^t [\delta(\tau+1) - 2\delta(\tau) + \delta(\tau-1)] d\tau$$

## Lec 24

### I. LTI systems & Complex Exponentials.

$$\begin{array}{c} x(t) = e^{st} \\ s \in \mathbb{C} \end{array} \rightarrow \boxed{h} \rightarrow ??$$

$$y(t) = (x * h)(t)$$

$$= \int_{-\infty}^{+\infty} h(\tau) \cdot x(t-\tau) d\tau$$

$$= \int_{-\infty}^{+\infty} h(\tau) \cdot e^{s(t-\tau)} d\tau$$

$$= e^{st} \int_{-\infty}^{+\infty} h(\tau) \cdot e^{-s\tau} d\tau$$

Define  $H(s) \triangleq \int_{t=-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau$

$$e^{st} \rightarrow \boxed{h} \rightarrow e^{st} \cdot H(s)$$

$\Rightarrow e^{st}$  is an eigenfunction of any LTI system.

$H(s)$  is called the Laplace Transform of  $h$ .

$$H: \mathbb{C} \rightarrow \mathbb{C}$$

$$s \mapsto H(s)$$

If  $s = j\omega$  where  $\omega \in \mathbb{R}$ ,

$$H(j\omega) = \int_{-\infty}^{+\infty} h(\tau) e^{-j\omega\tau} d\tau$$

$H(j\omega)$  is called the Fourier Transform of  $h$ .

As a Fourier Transform we think of  $H(j\omega)$  as a function of  $\omega$  (not as a function of  $j\omega$ ).

Example: Say  $x(t) = e^{-at} u(t)$  where  $a > 0$ .

$$X(j\omega) = \int_{-\infty}^{+\infty} x(\tau) \cdot e^{-j\omega\tau} d\tau$$

$$= \int_0^{\infty} e^{-a\tau} e^{-j\omega\tau} d\tau$$



$$\begin{aligned}
 &= \int_0^{\infty} e^{-(a+j\omega)s t} dt \\
 &= \left[ \frac{e^{-(a+j\omega)s t}}{-(a+j\omega)s} \right]_0^{\infty} \\
 &= \frac{1}{a+j\omega}
 \end{aligned}$$

### Remarks:

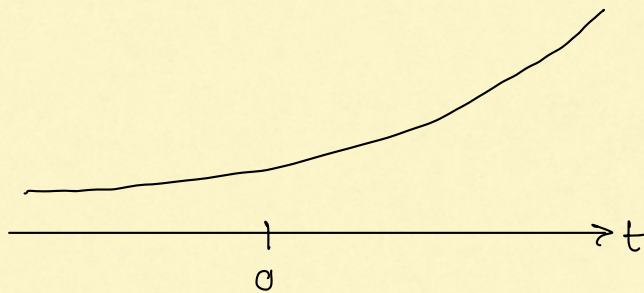
- For any linear transformation or a matrix  $A$  if  $Ax = \lambda x$  for some vector  $x$  and scalar  $\lambda$  then  $x$  is called an eigenvector of  $A$ .

Similarly, for an LTI system we have:

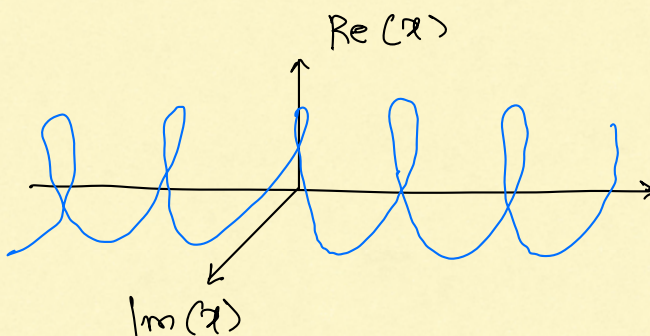
$$h * x = H(s) \cdot x \quad \text{if } x(t) = e^{st} \quad \forall t.$$

Hence  $e^{st}$  is an eigenfunction of any LTI system.

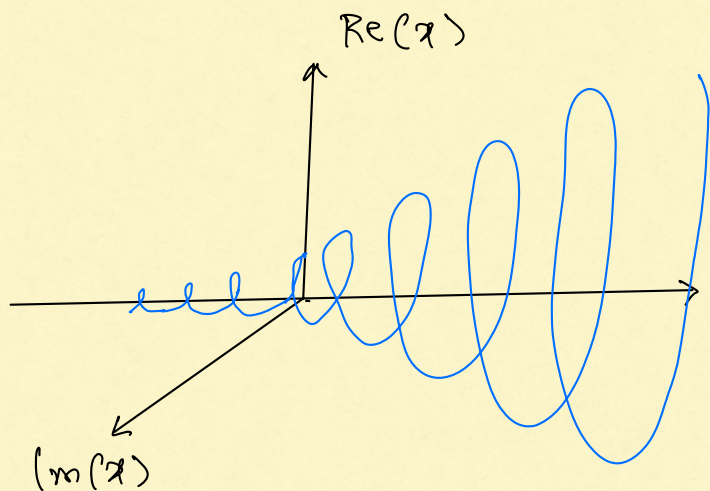
2.



$e^{st}$  if  $s$  is real  
and  $s > 0$ .



$e^{st}$  if  $s = j\omega$  where  
 $\omega$  is real.



$$\text{If } s = \sigma + j\omega$$

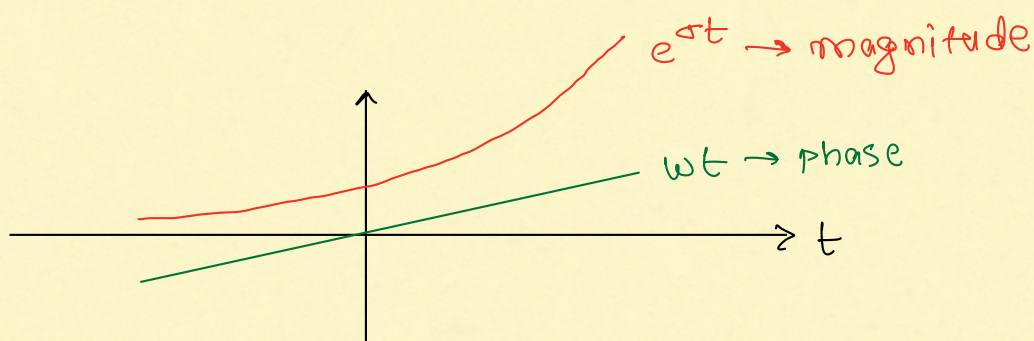
where  $\sigma > 0$  :

$$e^{st} = e^{(\sigma + j\omega)t}$$

$$= e^{\sigma t} \cdot e^{j\omega t}$$

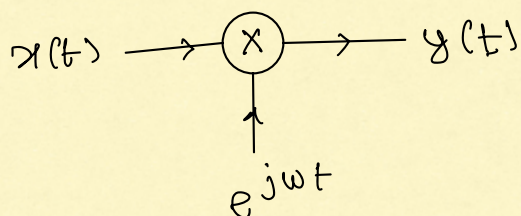
$$|e^{st}| = e^{\sigma t}$$

$$\angle e^{st} = \omega t$$



### 3. Complex Signals in Practice

Suppose we want to implement a system that performs:



where the input  $x$  is complex-valued.

In practice, we represent  $x(t)$  using two wires, one for real part and one for imaginary. Similarly for  $y(t)$  also.



